Dr. Madhu Srinivasan

Principal Research Staff · World Models · Generative Graphics · Efficient AI

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Summary

Principal Research Staff at AMD Research. Leads the team that built the company's **world-model research agenda** and its **real-time generative video model**. The team has filed 20 patent applications and works with 6 university labs. 17+ years in research and research leadership, including founding staff at KAUST's visualization lab and Lead Scientist at HPE Cray.

Work Experience

AMD Research

Austin, TX, USA

Principal Research Staff, Graphics and World Models

May 2022 - Present

- Lead the team of 4 researchers that built AMD's research agenda on world models and its implications for AMD hardware and software: consistent long-context generation, fine-grained control of camera, objects and scene elements, and real-time inference on small form-factor silicon.
- Demonstrated 130 fps streaming on Instinct GPUs using diffusion-forcing techniques with video foundation models, and a 75x speedup over the baseline on ROCm.
- Enabled LoRaQ, a data-free method for 4-bit post-training quantization of diffusion transformers with no loss of quality, completing a fully sub-16-bit pipeline (NeurIPS 2026).

HPE Cray

Seattle, WA, USA

Lead Scientist, Visualization and Machine Learning

September 2019 - April 2022

- Provide technical leadership within Cray's Alexandria Center of Excellence (COE). Analyze, understand, and visualize vast amounts of data proprietary to Cray's High Performance Computing systems. Act as the point of contact for the Alexandria Center for Excellence team throughout Cray.
- Leverage advanced Visualization and Machine Learning techniques to help Subject Matter Experts (SMEs) within Cray define, explore and predict systems behavior and failures. Conduct research on the data that Cray systems produce in order to increase system and application performance.
- Integrate open source analytics and AI projects, write application code, understand data science workflows, and troubleshoot and debug distributed systems. Manage multiple projects and tasks, prioritize work, and develop software projects on time.

KAUST Visualization Core Lab

Thuwal, Saudi Arabia

Lead Scientist, 3D Graphics and Visualization

July 2014 - July 2019

- Develop and implement strategic innovation within the lab in response to the University's Research and Industrial & Economic Development mission. Manage day-to-day operations and maintenance of large-scale interactive 3D Graphics and Virtual Reality facilities. Mentor and develop a team of graphics and data science experts. Collaborate with and provide support to the University's scientific communities and regional institutions.
- **Scientific Projects:** Enable scientific discovery through advanced 3D Visualization, and interactive graphics workflows in the areas of Clean Combustion Science, Marine Ecology, Earth and Climate Sciences, and Computational Fluid Dynamics (CFD). Collaborate with scientific users to create technology demonstrators for applying 3D graphics to large scale, complex and unstructured data. Follow agile project principles for providing flexible and responsive solutions to our users' unique scientific challenges.
- **Facilities:** Build and manage operations for large-scale visualization facilities such as Interactive Display Walls and Virtual Reality Environments. The facilities represent a capital investment of around $10 million, with an annual operational budget of $1 million.

KAUST Visualization Core Lab

Thuwal, Saudi Arabia

Visualization Scientist

April 2009 - June 2014

- **Scientific Projects:** Led visualization projects to build cutting-edge scientific decision-making tools for domain scientists and industry partners.
- **Software Development:** Architected solutions for large scale data visualization and interaction with distributed graphics on ultra-high resolution tiled-displays and virtual reality environments.

- **HPC Visualization:** Led efforts to design and implement *in situ* and streaming High Performance Computing (HPC) Visualization from supercomputing facilities at KAUST.

Skills

Areas of Expertise	Generative World Models, Diffusion Models, Efficient Inference, Real-Time Graphics, 3D Visualization, Scientific Data Analysis
Generative AI	Autoregressive Diffusion, Video Diffusion, Diffusion Forcing, Parameter-Efficient Fine-Tuning (LoRA), Multimodal Models, Quantization
ML Frameworks and Kernels	PyTorch, PyTorch Lightning, HuggingFace, Triton
Performance and Hardware	ROCm, AMD Instinct GPUs, Roofline Modeling, Performance Projection, Inference Optimization
Programming Languages	Python, Rust, Javascript, Elixir, C/C++
Data Visualization and 3D Graphics	D3.js, Three.js, Blender 3D, Unity3D Engine
Web Frameworks	Node.js, Phoenix, React, HTML5/WebGL, Observable Framework
Distributed and Network Computing	MPI, DisplayCluster, Erlang OTP

Selected Publications

- Vera: Decoupled Simulation for Efficient and Physically-Grounded World Modeling. Rishi Upadhyay, Aporva Amarnath, Sophie Yáng Shen, Achuta Kadambi, Mark Nutter, **Madhusudhanan Srinivasan**, ICLR (under review), 2027.
- Pareto LoRA: Mitigating Modality Imbalance in Unified Multimodal Models via Pareto-Optimal Gradient Integration. Xiwen Wei, Mark Nutter, **Madhusudhanan Srinivasan**, Radu Marculescu, WACV (accepted), 2027.
- trellis — An Analytics Framework for Understanding Slingshot Performance. **Madhu Srinivasan**, Dipanwita Mallick, Kristyn Maschhoff, Haripriya Ayyalasomayajula, Proceedings of the Cray User Group Conference, 2021.
- Insitu Workflow Specification. M. Parashar, P. Bremer, K. Heitmann, M. Larsen, J. Patchett, T. Peterka, **M. Srinivasan**, N. Röber, S. Frey, and B. Raffin, In Situ Visualization for Computational Science, Dagstuhl Report vol. 8, no. 7, p. 13, 2019.
- Software Complexity, Heterogeneity, and User-facing Issues. E. Bethel, J. Patchett, P. Messmer, M. Parashar, A. Bauer, **M. Srinivasan**, K. Ono, N. Röber, N. Gauger, J. Bennett, et al., In Situ Visualization for Computational Science, Dagstuhl Report vol. 8, no. 7, p. 34, 2019.
- RedSeaAtlas: A Visual Analytics Tool for Spatiotemporal Multivariate Red Sea Data. S. Afzal, S. Ghani, G. Tissington, S. Langodan, T. Jamil, **M. Srinivasan**, and I. Hoteit, Submitted to the Workshop on Visualization in Environmental Sciences 2019 (EnvirVis 2019), 2019.

Awards & Invited Talks

AMD Spotlight Award for Technical Leadership in World Models — *AMD*
Individual award — *2026*

AMD Fellows Forum: Company-wide Talk on World Models — *AMD*
AMD's annual internal conference — *June 2026*

AMD Spotlight Award for Communication Excellence — *AMD*
Individual award — *2023*

Ten-year Service Award — *KAUST, Saudi Arabia*
Recognition for Distinguished Service — *May 2019*

Dagstuhl Seminar: *In Situ* Visualization for Computational Science *Dagstuhl, Germany*

Invited Contributor & Speaker *July 2018*

- **Group Co-Leader:** *Workflow Specification Group*

Workshop on Visualization Infrastructure & Systems Technology *New Orleans, U.S.A*

Invited Speaker: KAUST Visualization Laboratory - Sustaining and Operating at Scale *November 2014*

The VISTech (Visualization Infrastructure & Systems Technology) workshop is organized as part of the Supercomputing Conference, and explores the intersection between human perception and large-scale visual analysis through the study of visualization interfaces and interactive displays. This rich intersection includes virtual reality systems, visualization through augmented reality, large-scale visualization systems, novel visualization interfaces, high-resolution interfaces, mobile displays, and visualization display middleware. The workshop provides a forum in which experts in the large-scale visualization technology field and users can come together to discuss state-of-the art technologies for visualization infrastructures and systems.

Selected Program Committees

2023 - Present	**Program Committee,** AMD Global Technical Authors Conference (GTAC), internal	*AMD*
2025	**Tutorial Chair,** Computer Vision and Pattern Recognition Conference (CVPR)	*Nashville, TN*
2019	**Program Committee,** EuroGraphics Symposium on Parallel Graphics and Visualization	*Porto, Portugal*
2016	**Program Committee,** IEEE Symposium on Large Data Analysis and Visualization	*Baltimore, MD*
2015	**Planning Committee,** Supercomputing Conference	*Austin, TX*

Education

Ph.D. Computer Science *Oregon, USA*

Oregon State University, College of Engineering *July 2005 - March 2009*

GPA 3.91/4.00

M.S. Computer Science *Oregon, USA*

Oregon State University, College of Engineering *September 2001 - June 2005*

GPA 3.90/4.00

B.Engg. Computer Science *Baroda, India*

Maharaja Sayajirao University, Faculty of Engineering and Technology *July 1997 - May 2001*

First Class with Distinction